

A Graph Attention LSTM Framework for Joint State-of-Charge and State-of-Health Estimation Using NASA and Real-World Electric Vehicle Data

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Abstract

Reliable estimation of State of Charge (SOC) and State of Health (SOH) is a fundamental requirement for safe and efficient lithium-ion battery management in electric vehicles (EVs), yet existing data-driven approaches often exhibit overfitting and limited generalization across heterogeneous operating conditions. To overcome these challenges, this study proposes a hybrid Graph Attention Long Short-Term Memory (GAT-LSTM) framework that simultaneously captures spatial interdependencies among battery variables and temporal degradation dynamics. The graph attention mechanism enables adaptive weighting of multi-dimensional feature interactions, while the LSTM component effectively models long-term aging behavior.

The proposed method is evaluated on the NASA prognostic datasets (B0005, B0007, B0018) and a large-scale real-time EV dataset comprising 40,000 samples with 20 operational features, including vehicle dynamics and DCDC characteristics. The model demonstrates strong predictive performance, achieving on the NASA dataset (SOH RMSE = 0.0155, MAE = 0.0121, $R^2 = 0.9667$; SOC RMSE = 0.0156, MAE = 0.0119, $R^2 = 0.9664$)

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and demonstrates superior performance on real-world data (SOH RMSE = 0.0058, MAE = 0.0040, $R^2 = 0.9998$; SOC RMSE = 0.2438, MAE = 0.1631, $R^2 = 0.9998$). Comparative results indicate that GAT-LSTM outperforms state-of-the-art baselines, including GCN-LSTM, Attention-LSTM, and Temporal Convolutional Networks, with significant error reductions. Ablation analysis further confirms the contribution of the graph attention mechanism. Moreover, minimal discrepancies between training, validation, and testing results demonstrate strong generalization across different battery aging stages. These findings establish the proposed framework as an effective and scalable solution for real-time battery management systems.

Keywords: State of Charge, State of Health, Lithium-ion batteries, Graph Attention Network, LSTM, Battery prognostics, Electric vehicles, Battery Management System

1. Introduction

Lithium-ion batteries (LIBs) have become a foundational technology for modern energy storage applications, including electric vehicles, hybrid systems, and renewable energy integration, owing to their high energy density, long lifespan, and operational efficiency [1, 2]. Despite these advantages, LIBs are inherently subject to complex degradation mechanisms driven by electrochemical, thermal, and mechanical interactions, which lead to gradual capacity fading, increased internal resistance, and potential safety risks during long-term operation [3, 4]. These degradation effects are further exacerbated under dynamic operating conditions such as high current rates and temperature variations, resulting in highly nonlinear battery behavior.

To ensure safe, reliable, and efficient operation, battery management systems (BMS) are employed to continuously monitor internal battery states and support control strategies. Among these, the state of charge (SOC) and state of health (SOH) are the most critical indicators, reflecting the available capacity and overall degradation status of the battery, respectively [5, 6]. Accurate estimation of SOC and SOH is essential for preventing overcharge and over-discharge, mitigating safety risks, and enabling optimal energy utilization. Furthermore, reliable state estimation plays a key role in facilitating predictive maintenance and supporting second-life applications of LIBs in energy storage systems. In this context, both publicly available datasets, such as NASA battery aging data, and proprietary electric vehicle datasets are

widely utilized to develop and validate advanced SOC and SOH prediction models, ensuring improved reliability and effective reuse of battery systems.

Accurate estimation of internal battery states remains a challenging task for battery management systems due to the complex and coupled dynamics of lithium-ion batteries. Among these states, the state of charge (SOC) and state of health (SOH) are the most critical, where SOC represents the real-time available capacity with fast time-varying characteristics, while SOH reflects the long-term degradation behavior with slow temporal dynamics. These two states are inherently coupled, as battery aging directly affects SOC estimation accuracy, while SOC operating conditions influence degradation progression [7, 8].

Conventional SOC estimation methods include current integration, open-circuit voltage (OCV), equivalent circuit model (ECM)-based approaches, and data-driven techniques. The current integration method suffers from error accumulation due to inaccurate initial conditions, while OCV-based methods require long rest periods, limiting their real-time applicability [9, 10]. ECM-based methods combined with filtering algorithms such as extended Kalman filter (EKF), unscented Kalman filter (UKF), and particle filter (PF) have been widely adopted due to their balance between computational efficiency and modeling capability [11, 12]. However, these approaches typically simplify battery dynamics and often neglect critical factors such as temperature variations, aging effects, and self-discharge, which can significantly degrade estimation accuracy under real-world conditions.

For SOH estimation, capacity degradation and internal resistance are commonly used indicators to quantify battery aging. Model-based approaches rely on accurate electrochemical or equivalent circuit representations, which are difficult to construct due to the nonlinear and time-varying nature of battery degradation. Although data-driven approaches, including deep learning models, have shown improved performance by capturing nonlinear relationships between measurable signals and SOH, they require large-scale labeled datasets and high computational resources [13, 14].

To address the coupling between SOC and SOH, joint estimation methods have gained increasing attention. Model-based joint estimation approaches, such as adaptive Kalman filtering combined with capacity identification, can improve estimation consistency but remain highly dependent on model accuracy [7]. Hybrid and data-driven joint estimation methods have demonstrated higher accuracy by leveraging rich operational data; however, they often involve extensive training processes and face challenges in real-time de-

ployment due to computational complexity and data dependency [15]. These limitations highlight the need for more robust and generalized approaches for accurate joint SOC and SOH estimation under diverse operating conditions. In recent years, data-driven approaches have gained significant attention for SOC and SOH estimation due to their strong capability to model complex nonlinear relationships directly from operational data without relying on explicit electrochemical models. Machine learning and deep learning techniques, including support vector machines, convolutional neural networks, and recurrent neural networks, have demonstrated promising performance in capturing battery degradation patterns and temporal dependencies [16, 17]. These approaches enable high-accuracy prediction by leveraging large-scale historical datasets and extracting latent degradation features that are difficult to model using traditional methods.

Despite their advantages, data-driven methods face several critical challenges. The performance of these models heavily depends on the quality and diversity of training data, which may not fully represent real-world operating conditions, leading to poor generalization across different battery chemistries and usage profiles [18]. Furthermore, feature extraction remains a key bottleneck, as identifying physically meaningful and degradation-relevant features from raw voltage, current, and temperature signals is nontrivial and often requires domain-specific expertise. Additionally, variations in operating conditions, such as temperature fluctuations, charge-discharge patterns, and measurement noise, introduce uncertainty and distribution shifts between training and deployment environments, significantly affecting prediction accuracy [19, 20].

To address these limitations, recent studies have focused on developing advanced deep learning frameworks that incorporate feature learning, temporal modeling, and domain adaptation strategies to enhance robustness and generalization. These advancements highlight the growing need for intelligent models capable of effectively handling complex spatio-temporal dependencies and real-world variability in battery systems. Recent advancements in deep learning have further improved SOC and SOH estimation by enabling more effective modeling of complex temporal dynamics and nonlinear degradation behaviors in lithium-ion batteries. In particular, architectures such as long short-term memory (LSTM) networks and gated recurrent units (GRU) have been widely adopted to capture sequential dependencies in battery data, leading to enhanced prediction accuracy under dynamic operating conditions [21, 22]. However, these models primarily focus on temporal cor-

relations while often overlooking the inherent structural relationships among multiple battery features and operating conditions.

To overcome this limitation, graph neural networks (GNNs) have recently emerged as a powerful framework for battery state estimation by modeling the interactions between different features as graph-structured data. By incorporating attention mechanisms, graph attention networks (GATs) can selectively learn the importance of different nodes and capture complex spatial dependencies, enabling more interpretable and robust predictions [23, 24]. The integration of GNNs with temporal models has shown promising potential in capturing spatio-temporal characteristics of battery degradation, thereby improving the accuracy of SOC and SOH estimation in complex scenarios.

Despite these advances, several challenges remain. Battery data are often affected by measurement noise, sensor inaccuracies, and operational variability, which can degrade model performance. Moreover, domain shifts caused by variations in battery types, aging conditions, and usage profiles limit the generalization capability of existing models [25]. Additionally, most current approaches lack effective mechanisms for handling anomalous data and adapting to unseen operating conditions, highlighting the need for more robust and generalized spatio-temporal learning frameworks for reliable battery state estimation.

To address the limitations of existing model-based and data-driven approaches, this work proposes a comprehensive framework for joint SOC and SOH estimation that integrates advanced feature learning, spatio-temporal modeling, and robust validation strategies. The proposed approach leverages a hybrid Graph Attention Network–Long Short-Term Memory (GAT-LSTM) architecture to effectively capture both spatial dependencies among multi-dimensional battery features and temporal dynamics across sequential operating conditions. Specifically, graph attention mechanisms are employed to model the complex interactions between voltage, current, and auxiliary variables, enabling adaptive feature weighting and improved representation learning, while the LSTM network captures long-term temporal degradation patterns for accurate state prediction.

To enhance robustness, the framework incorporates systematic preprocessing and feature normalization, ensuring stable learning across different operating regimes. Furthermore, a cross-battery validation strategy is adopted, where models are trained and evaluated on distinct battery units, enabling the assessment of generalization capability under varying degrada-

tion conditions. The model is jointly optimized using a multi-task learning objective that simultaneously minimizes SOC and SOH prediction errors, improving consistency and reducing error propagation between the two states.

Extensive experimental results demonstrate that the proposed method achieves high prediction accuracy for both SOC and SOH, with low mean absolute error (MAE), root mean square error (RMSE), and high coefficient of determination (R^2), indicating strong agreement between predicted and actual values. Additionally, the training and validation loss convergence behavior confirms stable learning and minimal overfitting, highlighting the robustness of the proposed framework. Overall, the integration of graph-based feature interaction modeling with temporal sequence learning provides a scalable and generalized solution for reliable battery state estimation under real-world operating conditions.

1.1. Main Contributions

The main contributions of this work are summarized as follows:

- A novel hybrid Graph Attention Network–Long Short-Term Memory (GAT-LSTM) framework is developed for joint estimation of state of charge (SOC) and state of health (SOH), enabling unified modeling of fast and slow battery dynamics.
- A dual-path graph-based feature learning mechanism is introduced to capture complex nonlinear interactions among multi-source battery signals, allowing adaptive feature importance weighting through attention mechanisms.
- A spatio-temporal learning strategy is designed by integrating graph attention with sequential modeling, effectively capturing both inter-feature dependencies and temporal degradation patterns.
- A cross-battery validation framework is employed to evaluate model generalization under varying operating conditions, demonstrating robustness across different battery units and usage profiles.
- Extensive experiments, including quantitative evaluation using MAE, RMSE, and R^2 metrics, along with training–validation convergence analysis, validate the effectiveness, stability, and practical applicability of the proposed method for real-time battery management systems.

1.2. Paper Organization

The remainder of this paper is organized as follows. Section 2 presents the proposed methodology, including the overall architecture and the GAT-LSTM-based spatio-temporal modeling framework. Section 3 describes the datasets, preprocessing steps, and experimental setup. Section 4 provides a comprehensive evaluation of the proposed method, including performance comparisons, ablation studies, and robustness analysis. Finally, Section 5 concludes the paper and outlines future research directions.

2. Proposed Methodology

To address the challenges of nonlinear battery dynamics, coupled state estimation, and limited generalization under varying operating conditions, this paper proposes a hybrid spatio-temporal learning framework for joint estimation of state of charge (SOC) and state of health (SOH) based on a Graph Attention Network–Long Short-Term Memory (GAT-LSTM) architecture, as illustrated in Fig. 1. The overall process is structured as follows. First, raw battery measurements, including voltage, current, and auxiliary operational signals, are preprocessed and normalized to ensure stable learning across different scales and operating regimes. The input features are then organized into sequential time-series segments and complementary cycle-related representations, capturing both instantaneous and long-term degradation characteristics.

Next, the temporal sequences are transformed into graph-structured data by constructing sequential graph connections, where each time step is treated as a node and edges represent temporal adjacency, enabling structured representation of feature interactions. The graph-based feature sequences are then processed through parallel GAT layers, which learn adaptive attention weights to capture complex interdependencies among multi-dimensional battery variables. Subsequently, the extracted spatial features are fused and passed to an LSTM network to model long-term temporal dependencies and degradation evolution patterns.

Finally, the learned spatio-temporal representations are fed into task-specific fully connected layers to simultaneously predict SOC and SOH in a multi-task learning framework. The model is trained using a joint loss function that minimizes the prediction errors of both states, thereby enhancing estimation consistency and reducing error propagation. This integrated approach enables robust and accurate battery state estimation under dynamic

and real-world operating conditions.. The overall architecture of the proposed model is illustrated in Fig. 1, which integrates graph-based feature learning with temporal sequence modeling.

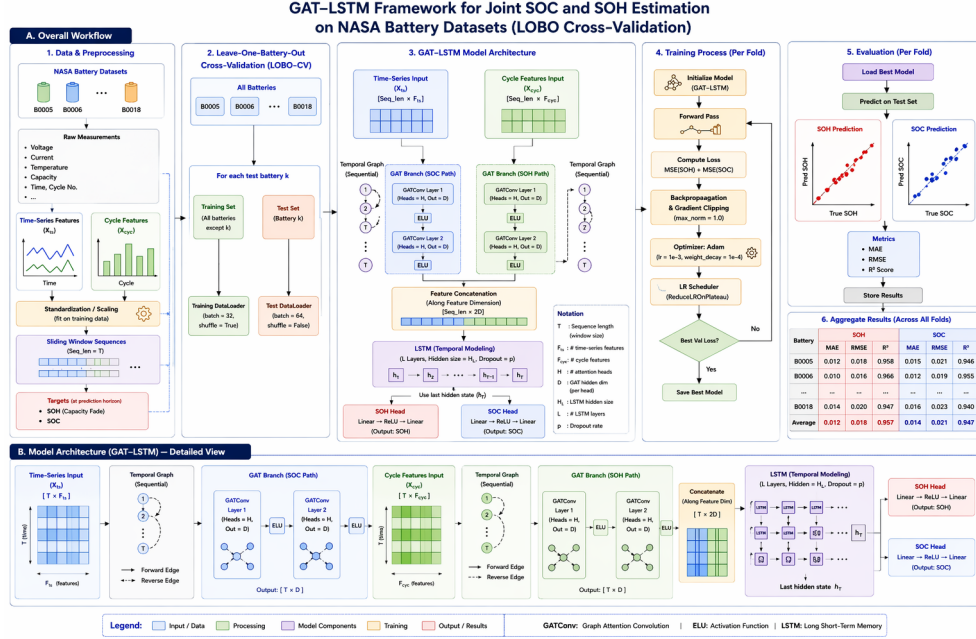


Figure 1: Proposed GAT-LSTM architecture for joint SOC and SOH estimation.

2.1. Data Preprocessing

The SOC and SOH estimation process begins with the acquisition of multivariate operational data from the battery management system (BMS), including voltage, current, and auxiliary driving-related signals. These measurements are recorded under real-world operating conditions and inherently exhibit noise, scale variations, and nonlinear dependencies, which can adversely affect model performance if not properly handled.

To ensure stable and efficient learning, this work employs a structured preprocessing pipeline consisting of feature selection, normalization, and sequence construction. Specifically, two distinct feature groups are defined to capture complementary battery characteristics. The first group corresponds to SOC-related features, including vehicle speed, distance, acceleration, voltage, and current, which primarily reflect short-term dynamic behavior. The

second group consists of SOH-related features, such as charge/discharge capacity, available capacity, and electrical measurements, which encode long-term degradation information. This dual-feature representation enables the model to simultaneously learn fast-varying and slow-varying battery dynamics.

To eliminate the effect of scale differences among features and improve numerical stability, z-score normalization is applied to both input features and target variables. The standardized feature $\tilde{x}$ is computed as:

$$\tilde{x} = \frac{x - \mu}{\sigma} \quad (1)$$

where x denotes the original feature value, μ and σ represent the mean and standard deviation of the corresponding feature, respectively. This transformation ensures that all features contribute equally during training and prevents dominance of high-magnitude variables.

Following normalization, the time-series data are segmented into fixed-length sequences using a sliding window approach. For each sample, a sequence of length T is constructed as:

$$X_t = [x_t, x_{t+1}, \dots, x_{t+T-1}] \quad (2)$$

where T denotes the sequence length. The corresponding SOC and SOH targets are defined at the next time step, enabling supervised learning of temporal dependencies.

To further facilitate graph-based modeling, each sequence is represented as a structured graph, where individual time steps correspond to nodes and edges encode temporal adjacency between consecutive samples. This representation allows the model to capture both sequential and relational dependencies in battery data. The resulting preprocessed dataset is then used as input to the proposed GAT-LSTM framework for joint SOC and SOH estimation.

Let the input dataset be represented as:

$$\mathbf{X} = \{x_1, x_2, \dots, x_T\}, \quad x_t \in \mathbb{R}^d \quad (3)$$

where T denotes the number of time steps and d represents the number of features including voltage, current, temperature, and operational parameters.

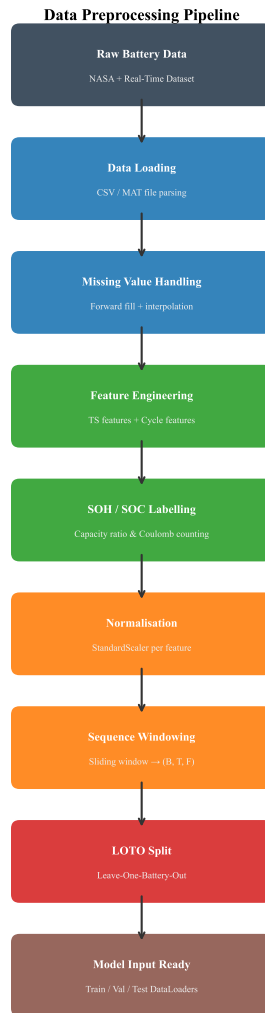


Figure 2: Data preprocessing pipeline including normalization, filtering, and feature extraction.

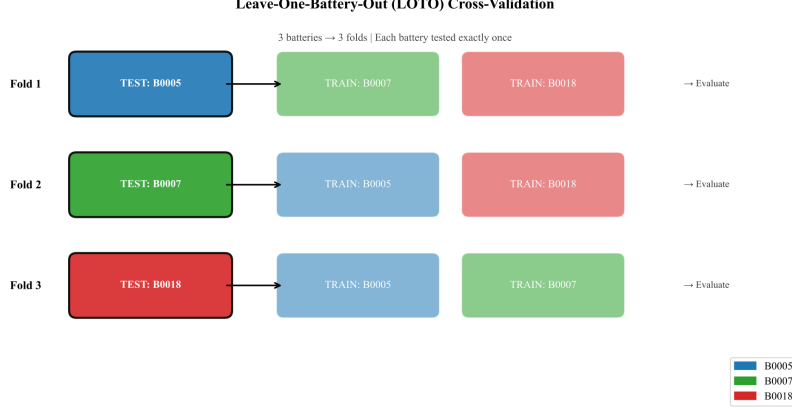


Figure 3: Leave-One-Battery-Out (LOTO) cross-validation strategy. Three folds are used, with each battery (B0005, B0007, B0018) serving as the test set exactly once while the remaining two batteries are used for training.

2.2. Graph-Structured Representation and GAT-Based Feature Learning

To effectively capture the structural relationships and temporal interactions within battery data, the preprocessed sequential inputs are transformed into graph-structured representations and processed using Graph Attention Networks (GAT). This approach enables adaptive learning of feature importance and interaction patterns across time steps.

In this work, a fully connected temporal graph structure is adopted, allowing each time step to interact with all other time steps within the sequence. This design enables the model to capture global temporal dependencies and learn complex interactions across the entire sequence through attention mechanisms.

2.2.1. Graph Construction

Given a normalized sequence of length T , the node feature matrix is defined as:

$$X = [x_1, x_2, \dots, x_T] \in \mathbb{R}^{T \times F} \quad (4)$$

where $x_t \in \mathbb{R}^F$ represents the feature vector at time step t , and F denotes the number of input features. In this work, separate feature sets are constructed for SOC-related and SOH-related variables, forming two parallel input streams.

The graph structure is defined by connecting temporally adjacent nodes. Specifically, bidirectional edges are established as:

$$E = \{(t, t+1), (t+1, t)\}, \quad t = 1, 2, \dots, T \quad (5)$$

To preserve continuity, cyclic connections are introduced such that $(T, 1)$ and $(1, T)$ are also included. This results in a structured temporal graph where each node interacts with its immediate neighbors.

2.2.2. Graph Attention Mechanism

Given the constructed graph, node features are updated using the Graph Attention Network. For each node i , a linear transformation is first applied:

$$h_i = Wx_i \quad (6)$$

where W is a learnable weight matrix.

The attention coefficient between node i and its neighbor j is computed as:

$$e_{ij} = \text{LeakyReLU}(a^T[h_i \parallel h_j]) \quad (7)$$

where a is the learnable attention vector and $\parallel$ denotes concatenation.

These coefficients are normalized using the softmax function:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})} \quad (8)$$

where $\mathcal{N}(i)$ denotes the neighborhood of node i .

The updated node representation is then obtained as:

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} h_j \right) \quad (9)$$

where $\sigma(\cdot)$ is a nonlinear activation function.

2.2.3. Multi-Head Attention and Feature Extraction

To improve model stability and representation capacity, multi-head attention is employed. Specifically, K independent attention heads are applied and their outputs are concatenated:

$$h'_i = \parallel_{k=1}^K \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(k)} W^{(k)} x_j \right) \quad (10)$$

In this work, multiple GAT layers are stacked to progressively refine feature representations. The first layer captures low-level feature interactions, while subsequent layers extract higher-level abstractions. Separate GAT modules are applied to SOC-related and SOH-related feature streams, enabling the model to learn distinct spatial relationships corresponding to short-term dynamics and long-term degradation behavior.

For batch processing, multiple graph sequences are combined using an index-shifting strategy to construct a block-diagonal adjacency structure, ensuring computational efficiency while preserving independence between samples.

The output of the GAT layers provides enriched node-level feature representations that encode both temporal adjacency and inter-feature relationships. These representations are subsequently used as input to the temporal modeling stage described in the next section.

2.3. Temporal Modeling Using LSTM

While the Graph Attention Network (GAT) effectively captures structural relationships and feature interactions, it does not explicitly model long-term temporal dependencies inherent in battery degradation processes. To address this limitation, a Long Short-Term Memory (LSTM) network is employed to learn sequential patterns and temporal evolution of battery states.

2.3.1. Feature Fusion

The output features from the parallel GAT branches corresponding to SOC-related and SOH-related inputs are first fused to form a unified spatio-temporal representation. Let $H^{\text{SOC}} \in \mathbb{R}^{T \times d}$ and $H^{\text{SOH}} \in \mathbb{R}^{T \times d}$ denote the GAT-extracted features for SOC and SOH streams, respectively. The fused feature representation is obtained via concatenation:

$$H = [H^{\text{SOC}} \parallel H^{\text{SOH}}] \in \mathbb{R}^{T \times 2d} \quad (11)$$

where $\parallel$ denotes the concatenation operation.

2.3.2. LSTM-Based Temporal Learning

The fused feature sequence H is then fed into the LSTM network to model temporal dependencies. At each time step t , the LSTM updates its internal

states according to:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (12)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (13)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (14)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (15)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (16)$$

$$h_t = o_t \odot \tanh(c_t) \quad (17)$$

where f_t , i_t , and o_t denote the forget, input, and output gates, respectively; c_t is the cell state; h_t is the hidden state; $\sigma(\cdot)$ represents the sigmoid activation function; and $\odot$ denotes element-wise multiplication.

The LSTM effectively captures both short-term fluctuations and long-term degradation trends in battery behavior by selectively retaining or discarding information through its gating mechanism.

2.3.3. Joint SOC and SOH Prediction

After temporal modeling, the final hidden state of the LSTM, denoted as h_T , is used as a compact representation of the entire input sequence. This representation is then passed through two separate fully connected layers to simultaneously predict SOC and SOH:

$$\hat{y}_{\text{SOC}} = f_{\text{SOC}}(h_T) \quad (18)$$

$$\hat{y}_{\text{SOH}} = f_{\text{SOH}}(h_T) \quad (19)$$

where $f_{\text{SOC}}(\cdot)$ and $f_{\text{SOH}}(\cdot)$ denote task-specific regression heads.

This multi-task learning formulation enables the model to jointly learn SOC and SOH, allowing shared feature representations to improve prediction accuracy while reducing error propagation between the two states. The integration of GAT-based spatial feature extraction with LSTM-based temporal modeling provides a unified spatio-temporal framework for robust battery state estimation.

2.3.4. SOC Estimation Model

The state of charge (SOC) estimation is formulated as a regression task based on the spatio-temporal features extracted from the proposed GAT-LSTM framework. The Graph Attention Network (GAT) is first employed to capture complex interactions among input features across temporal nodes.

Unlike conventional convolutional neural networks, GAT dynamically computes attention coefficients between connected nodes, enabling adaptive aggregation of relevant information.

For each node i , the input feature vector x_i is linearly transformed as:

$$h_i = Wx_i \quad (20)$$

where W denotes the learnable weight matrix.

The attention coefficient between node i and its neighboring node j is computed as:

$$e_{ij} = \text{LeakyReLU} \left(a^T [h_i || h_j] \right) \quad (21)$$

where a is the learnable attention vector.

The normalized attention weights are obtained using:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})} \quad (22)$$

The updated node representation is then calculated as:

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} h_j \right) \quad (23)$$

To enhance representation capability, multi-head attention is applied:

$$h'_i = \bigoplus_{k=1}^K \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(k)} W^{(k)} x_j \right) \quad (24)$$

The resulting spatial features are then passed to the LSTM network to model temporal dependencies. The final hidden state h_T is used to estimate SOC through a fully connected regression layer:

$$\hat{y}_{\text{SOC}} = f_{\text{SOC}}(h_T) \quad (25)$$

This formulation enables accurate tracking of fast-varying SOC dynamics by leveraging both spatial feature interactions and temporal evolution patterns.

2.3.5. SOH Estimation Model

The state of health (SOH) estimation is modeled as a long-term degradation prediction task using the same unified GAT-LSTM framework. While SOC reflects short-term dynamics, SOH captures slow-varying aging characteristics, which require effective modeling of long-range temporal dependencies.

The SOH-related input features, including capacity and degradation indicators, are first processed through dedicated GAT layers to extract spatial correlations among feature variables. The attention mechanism allows the model to emphasize degradation-sensitive features while suppressing less informative signals, leading to improved feature representation.

The extracted spatial features are then fused with SOC-related features and fed into the LSTM network. Through its gating mechanism, the LSTM selectively retains long-term degradation information and filters out irrelevant variations. The final hidden state encodes the overall degradation trajectory of the battery.

The SOH estimation is obtained via a task-specific regression head:

$$\hat{y}_{\text{SOH}} = f_{\text{SOH}}(h_T) \quad (26)$$

By jointly learning SOC and SOH within a shared spatio-temporal framework, the model effectively captures the intrinsic coupling between short-term operational behavior and long-term degradation processes. This multi-task formulation improves estimation accuracy and enhances model robustness under varying operating conditions.

2.4. Loss Function and Training Strategy

To achieve accurate and consistent joint estimation of SOC and SOH, the proposed model is trained using a multi-task learning objective that simultaneously minimizes the prediction errors of both states. Specifically, the loss function is defined as the sum of the mean squared error (MSE) losses for SOC and SOH:

$$\mathcal{L} = \mathcal{L}_{\text{SOC}} + \mathcal{L}_{\text{SOH}} \quad (27)$$

where

$$\mathcal{L}_{\text{SOC}} = \frac{1}{N} \sum_{i=1}^N \left(y_{\text{SOC}}^{(i)} - \hat{y}_{\text{SOC}}^{(i)} \right)^2 \quad (28)$$

$$\mathcal{L}_{SOH} = \frac{1}{N} \sum_{i=1}^N \left(y_{SOH}^{(i)} - \hat{y}_{SOH}^{(i)} \right)^2 \quad (29)$$

Here, y_{SOC} and y_{SOH} denote the ground truth values, while $\hat{y}_{SOC}$ and $\hat{y}_{SOH}$ represent the predicted outputs. This joint optimization encourages the model to learn shared representations that benefit both tasks while reducing error propagation between SOC and SOH.

The model parameters are optimized using the Adam optimizer due to its efficiency in handling nonlinear and high-dimensional optimization problems. An adaptive learning rate scheduling strategy is employed using a Reduce-on-Plateau mechanism, which decreases the learning rate when the validation loss stagnates, thereby improving convergence stability.

To enhance training robustness and prevent gradient explosion, gradient clipping is applied during backpropagation, constraining the gradient norm within a predefined threshold. Dropout regularization is also incorporated within the network to mitigate overfitting and improve generalization performance.

Furthermore, a cross-battery validation strategy is adopted to evaluate the generalization capability of the proposed model. In this approach, data from one battery unit are used as the test set, while the remaining data are used for training. This process is repeated across all battery units, ensuring that the model is validated under varying operating conditions and degradation patterns.

The model performance is evaluated using standard regression metrics, including mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R^2), which provide a comprehensive assessment of prediction accuracy and robustness. This training framework ensures stable convergence, strong generalization, and reliable performance in real-world battery management applications.

3. Experimental Setup and Datasets

3.1. Dataset

In this study, a real-world battery dataset comprising approximately 40,000 samples with 20 operational features is utilized to evaluate the performance of the proposed GAT-LSTM framework. The dataset contains multivariate measurements collected from battery management systems (BMS) under dynamic operating conditions, reflecting practical usage scenarios.

The recorded features include electrical and operational variables such as battery voltage, battery current, charge capacity, discharge capacity, available capacity, vehicle speed, distance, and acceleration signals. These features capture both instantaneous operational behavior and long-term degradation characteristics of the battery. In particular, battery voltage and current provide critical information about real-time electrochemical behavior, while capacity-related features encode degradation trends essential for SOH estimation.

The state of charge (SOC) is obtained directly from the dataset as a measured or reference value, while the state of health (SOH) is derived using capacity degradation indicators. Specifically, SOH is defined based on capacity fade as:

$$\text{SOH} = \frac{C_{\text{current}}}{C_{\text{rated}}} \quad (30)$$

where C_{current} represents the available capacity at a given time, and C_{rated} denotes the nominal capacity of the battery.

The dataset exhibits significant variability in operating conditions, including fluctuations in load profiles, driving patterns, and environmental factors. Such variability introduces nonlinearities and noise into the data, making it well-suited for evaluating the robustness and generalization capability of data-driven models.

For model development, the dataset is organized into multiple battery groups, enabling cross-battery validation. In this setup, data from one battery unit are used as the test set, while the remaining units are used for training, ensuring that the model is evaluated on unseen battery conditions. This strategy provides a realistic assessment of the model’s ability to generalize across different degradation patterns and operating environments.

To further illustrate the degradation characteristics of lithium-ion batteries, Fig. 4 presents the capacity fade trajectories of representative NASA battery cells (B0005, B0007, and B0018). It can be observed that the degradation process exhibits strong nonlinearity, with relatively slow capacity decay in early cycles followed by accelerated degradation in later stages. Additionally, noticeable variability exists across different cells, reflecting differences in aging patterns and operating conditions.

Such nonlinear and cell-dependent degradation behaviors pose significant challenges for accurate SOH estimation and highlight the necessity of robust data-driven models capable of generalizing across diverse battery conditions.

Therefore, these datasets serve as reliable benchmarks for validating battery degradation modeling and state estimation methods [26, 27].

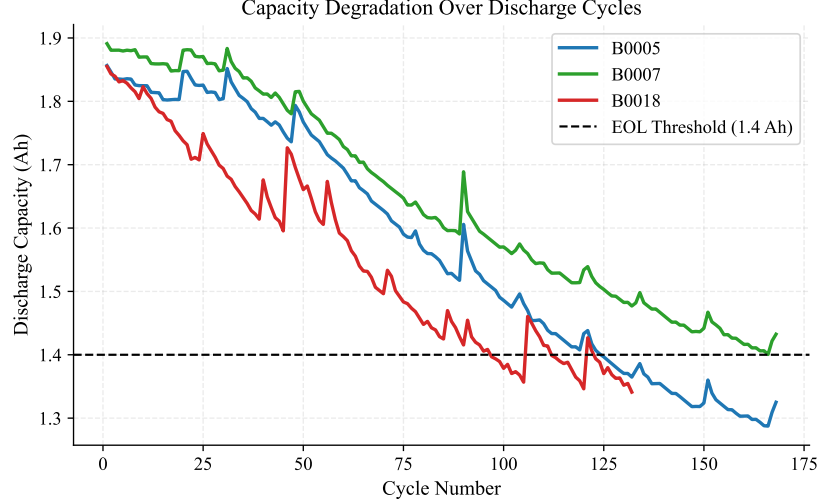


Figure 4: Capacity degradation curves of NASA batteries (B0005, B0007, B0018).

To ensure fair evaluation and minimize temporal bias, the dataset is carefully partitioned into training, validation, and testing subsets. Although a sliding window approach is used for sequence generation, care is taken to ensure that overlapping sequences do not introduce unintended information leakage between different subsets. This setup reflects realistic prediction conditions and prevents overly optimistic performance estimates.

3.2. Real-Time EV Dataset

In addition to benchmark datasets, a real-world electric vehicle (EV) dataset comprising approximately 40,000 samples is utilized to evaluate the robustness of the proposed model under practical operating conditions. The dataset was provided through an industrial collaboration with Adani New Industries Ltd., Gujarat, India, and includes multiple operational features such as battery voltage, current, temperature, vehicle speed, and DC-DC converter characteristics.

The dataset captures dynamic driving conditions, including acceleration, deceleration, and varying load profiles, introducing significant nonlinearities and noise typical of real-world environments. Such variability makes accurate

state estimation more challenging and highlights the importance of robust spatio-temporal modeling.

Fig. 5 illustrates the feature correlation matrix of the EV dataset, revealing the interdependencies among key variables used for SOC and SOH estimation.

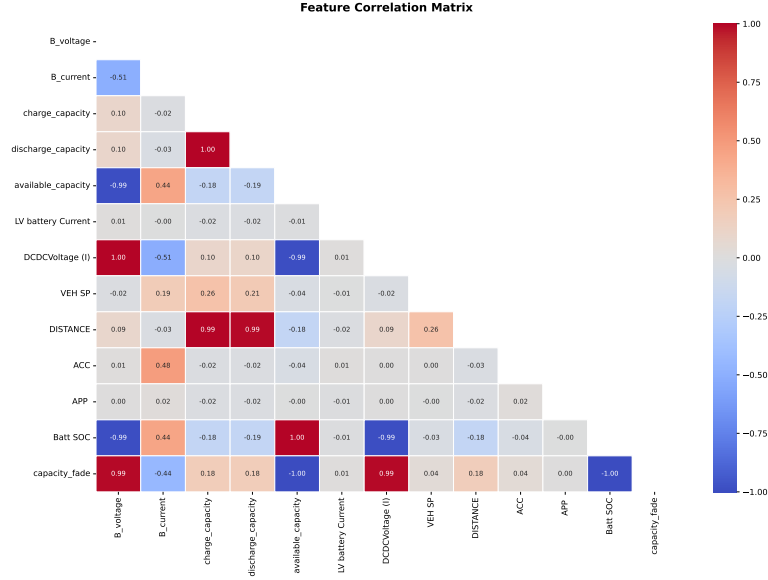


Figure 5: Feature correlation matrix of the real-time EV dataset. The heatmap reveals the interdependencies among key variables, where battery voltage and current exhibit strong correlations with SOC and SOH, justifying their selection as primary inputs for the proposed model.

In addition to feature-level analysis, the temporal behavior of key battery signals is illustrated in Fig. 6. The figure presents voltage, current, SOC, and SOH variations under dynamic operating conditions. It can be observed that the current exhibits highly fluctuating behavior due to varying load demands, while the voltage remains relatively stable with minor transients. The SOC shows gradual depletion over time, and the SOH reflects a slow degradation trend, consistent with battery aging characteristics.

These observations highlight the nonlinear and time-dependent nature of battery behavior in real-world scenarios, reinforcing the need for advanced spatio-temporal models such as the proposed GAT-LSTM framework.

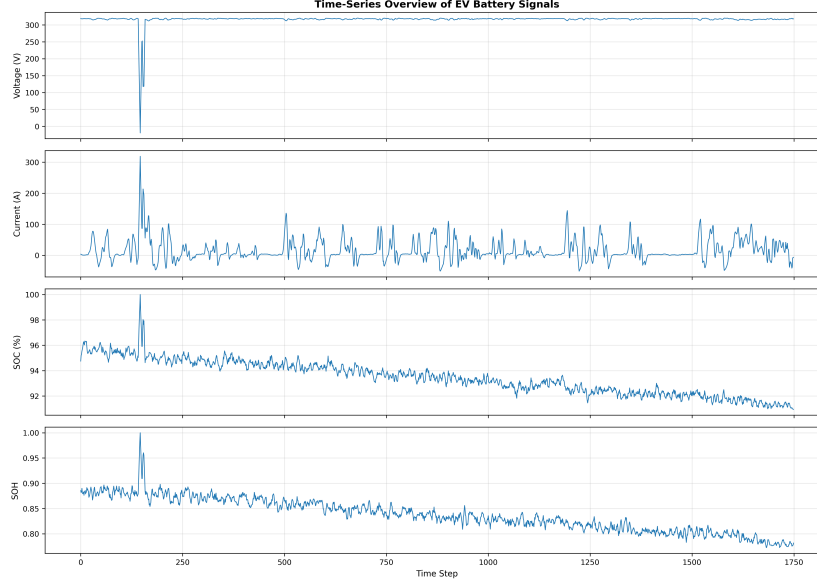


Figure 6: Time-series visualization of EV battery signals, including voltage, current, SOC, and SOH under dynamic operating conditions.

3.3. Model Parameter Settings

This section presents the key hyperparameters and training configurations used in the proposed GAT-LSTM framework. The model is implemented using the PyTorch deep learning library, and all experiments are conducted under a GPU-enabled environment to ensure efficient training.

The input data are segmented into fixed-length sequences using a sliding window approach, with a sequence length of 50. To improve convergence stability, all input features and target variables are normalized prior to training.

The model architecture consists of two Graph Attention Network (GAT) layers followed by a Long Short-Term Memory (LSTM) network. The GAT layers are used to capture feature interactions, while the LSTM models temporal dependencies. The hidden dimension of the GAT layers is set to 64, and multi-head attention with 4 heads is employed to enhance feature representation. The LSTM network consists of 2 layers with 128 hidden units.

The model is trained using the Adam optimizer with an initial learning rate of 1×10^{-3} and a weight decay of 1×10^{-4} . A ReduceLROnPlateau scheduler is used to adaptively adjust the learning rate during training. The batch size is set to 32, and the model is trained for 100 epochs. Dropout with

a rate of 0.3 is applied to prevent overfitting. The complete hyperparameter configuration is summarized in Table 1.

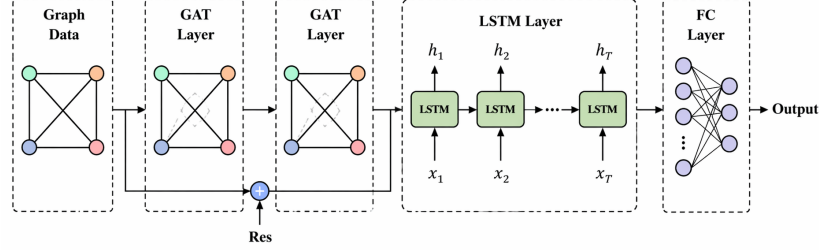


Figure 4. Overall structure of the proposed GAT-LSTM model.

Figure 7: Overall sturcture of the proposed GAT-LSTM framework for joint SOC and SOH estimation.

The overall structure of the proposed GAT-LSTM framework is illustrated in Fig. 7. The model consists of parallel Graph Attention Network (GAT) layers for feature interaction learning, followed by a Long Short-Term Memory (LSTM) network for temporal modeling. The extracted spatio-temporal features are then passed through fully connected layers to jointly estimate SOC and SOH.

3.4. Baseline Models

To validate the effectiveness of the proposed GAT-LSTM framework, several representative baseline models are implemented for comparison, including GCN-LSTM, Attention-LSTM, and Temporal Convolutional Network (TCN).

The GCN-LSTM model replaces the graph attention mechanism with graph convolutional layers to capture feature relationships, followed by an LSTM network for temporal modeling. The Attention-LSTM model employs a standard LSTM architecture enhanced with an attention mechanism to focus on important temporal features. The TCN model utilizes dilated causal convolutions to model temporal dependencies without recurrent structures.

For a fair comparison, all baseline models are trained using the same input features, data preprocessing pipeline, train-test splits, and evaluation metrics as the proposed method. Hyperparameters such as learning rate, batch size, and training epochs are kept consistent across models, while model-specific

Table 1: Hyperparameter configuration of the proposed GAT-LSTM model.

Hyperparameter	Value	Description
Framework	PyTorch 2.0 + PyTorch Geometric	Implementation platform
GAT Layers	2	Number of GATConv layers
Attention Heads	4	Multi-head attention
GAT Hidden Dim	64	Per-head dimension
LSTM Layers	2 (Bidirectional)	Temporal modeling depth
LSTM Hidden Size	128	Hidden units per direction
Fusion MLP	320→128→64	Fully connected layers
Dropout Rate	0.3	Regularization
Optimizer	AdamW	Weight decay = 1×10^{-4}
Learning Rate	0.001	Initial value
LR Scheduler	ReduceLROnPlateau	Factor=0.5, Patience=10
Batch Size	32	Training batch size
Epochs	50	Maximum training epochs
Early Stopping	15	Patience epochs
Sequence Length	50	Sliding window size
Gradient Clip	1.0	Maximum gradient norm
NASA Split	LOTO	Leave-One-Battery-Out
RT Split	80-10-10	Train-Val-Test (%)
Loss Function	MSE(SOH) + MSE(SOC)	Combined loss

parameters are selected based on commonly adopted configurations in the literature. All models are implemented in the same experimental environment to ensure reproducibility and unbiased comparison.

3.5. Evaluation Metrics

To quantitatively evaluate the performance of the proposed model, three standard regression metrics are employed: mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R^2). These metrics are used to assess the accuracy of both SOC and SOH predictions.

The evaluation metrics are defined as follows:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (31)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (32)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (33)$$

where N denotes the total number of samples, y_i and $\hat{y}_i$ represent the ground truth and predicted values, respectively, and $\bar{y}$ denotes the mean of the ground truth values.

These metrics provide complementary insights into model performance, where MAE measures average prediction error, RMSE emphasizes larger deviations, and R^2 evaluates the goodness of fit between predicted and actual values.

4. Results and Discussion

This section presents the performance evaluation of the proposed GAT-LSTM model for SOC and SOH estimation. The results are analyzed in terms of training behavior, prediction accuracy, comparative performance, and error characteristics across both NASA and real-time datasets.

4.1. Training Convergence Analysis

The training and validation loss curves of the proposed model are shown in Fig. 8. The model exhibits stable convergence with minimal gap between training and validation loss, indicating effective learning and reduced overfitting.

4.2. SOC and SOH Prediction Performance

The prediction performance of the proposed model is evaluated by comparing predicted and actual values of SOC and SOH. Fig. 9 and Fig. 10 illustrate the close alignment between predicted and ground truth values, demonstrating the high accuracy of the model.

Table 2: Per-battery performance of GAT-LSTM on NASA dataset.

Battery	SOH MAE	SOH RMSE	SOH R^2	SOC MAE	SOC RMSE	SOC R^2
B0005	0.0147	0.0179	0.9692	0.0142	0.0176	0.9705
B0007	0.0064	0.0085	0.9900	0.0061	0.0087	0.9896
B0018	0.0153	0.0202	0.9408	0.0155	0.0205	0.9391
Average	0.0121	0.0155	0.9667	0.0119	0.0156	0.9664

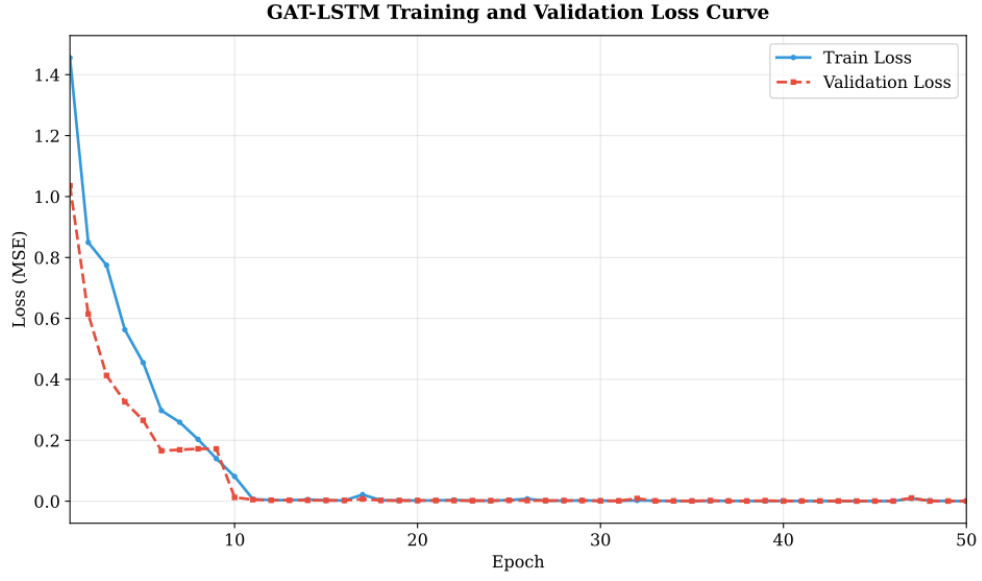


Figure 8: Training and validation loss convergence of the proposed GAT-LSTM model.

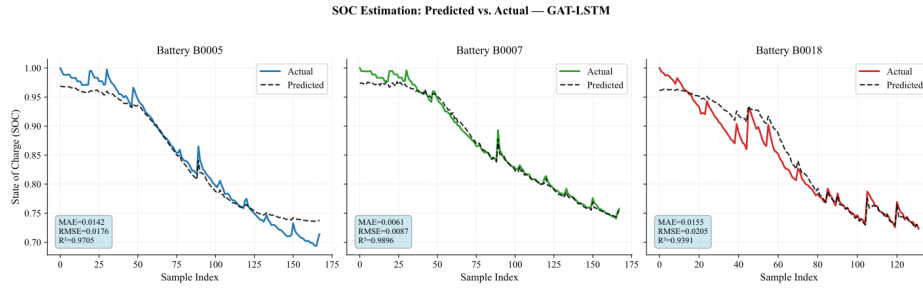


Figure 9: SOC prediction: actual vs predicted values.

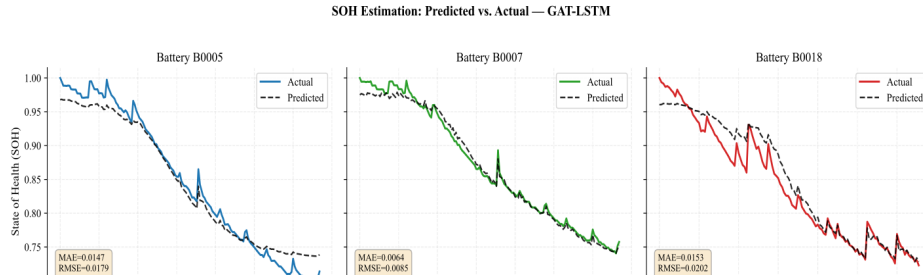


Figure 10: SOH prediction: actual vs predicted values.

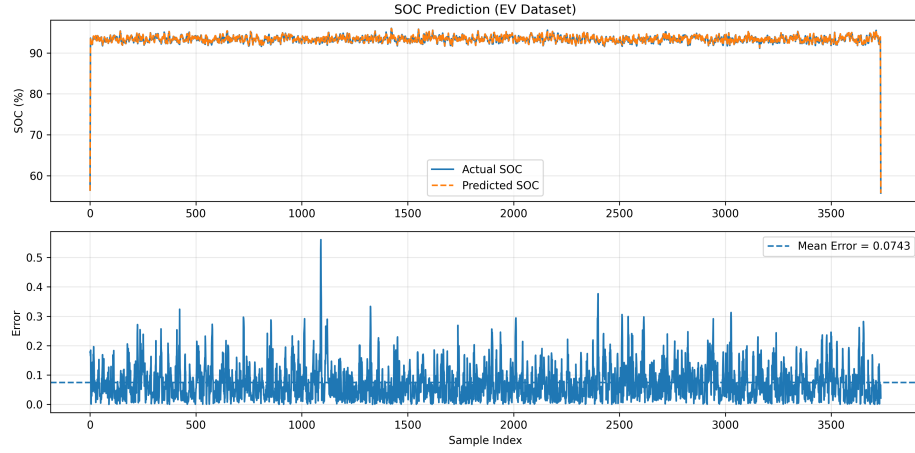


Figure 11: SOC estimation results on the real-time EV dataset. The predicted values closely follow the ground truth, demonstrating accurate estimation under dynamic operating conditions.

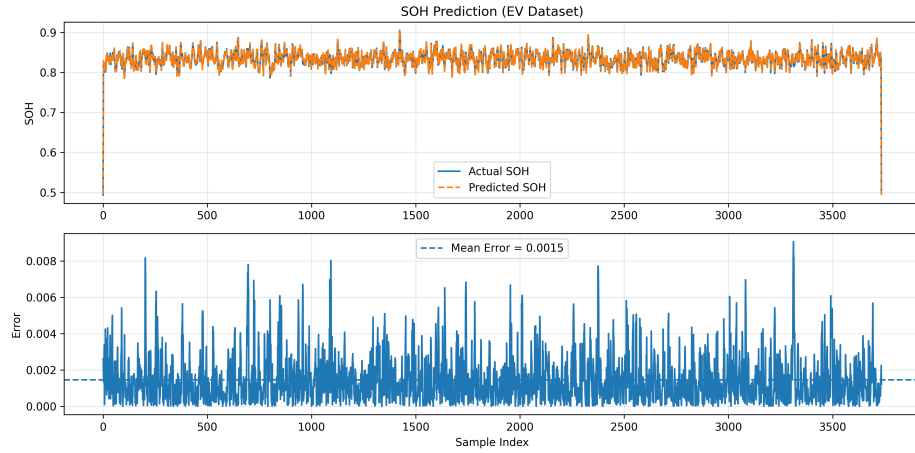


Figure 12: SOH estimation results on the real-time EV dataset, showing effective tracking of battery degradation with low prediction error.

In addition to the NASA benchmark dataset, the proposed model is further evaluated on a real-time EV dataset. Fig. 11 and Fig. 12 illustrate the SOC and SOH estimation results, respectively. It can be observed that the predicted values closely follow the ground truth with minimal deviation, demonstrating the robustness and generalization capability of the proposed model under real-world operating conditions characterized by noise and dynamic variability.

4.3. Regression Analysis

To further validate the prediction accuracy, regression plots are presented in Fig. 13. The results show strong linear correlation between predicted and actual values, confirming the reliability of the proposed model.

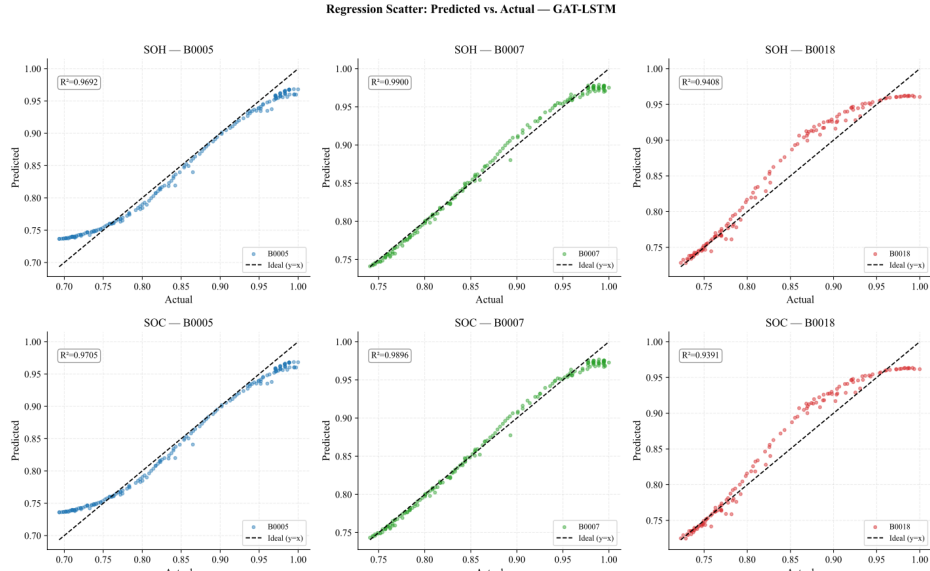


Figure 13: Regression analysis of predicted vs actual values.

In this work, a fully connected temporal graph structure is adopted, allowing each time step to interact with all other time steps within the sequence. This design enables the model to capture global temporal dependencies and learn complex interactions across the entire sequence through attention mechanisms. It is important to note that SOC exhibits relatively smooth variations under normal operating conditions, which contributes to higher prediction accuracy compared to more stochastic signals.

Table 3: Performance comparison on NASA dataset.

Model	SOH MAE	SOH RMSE	SOH R^2	SOC MAE	SOC RMSE	SOC R^2
GAT-LSTM	0.0121	0.0155	0.9667	0.0119	0.0156	0.9664
GCN-LSTM	0.0184	0.0214	0.9378	0.0176	0.0207	0.9400
Attention-LSTM	0.0151	0.0186	0.9517	0.0159	0.0195	0.9495
TCN	0.0287	0.0310	0.8741	0.0192	0.0217	0.9404

4.4. Model Comparison

The performance of the proposed GAT-LSTM model is compared with state-of-the-art models, including GCN-LSTM, Attention-LSTM, and Temporal Convolutional Networks (TCN). The comparison results are shown in Fig. 14.

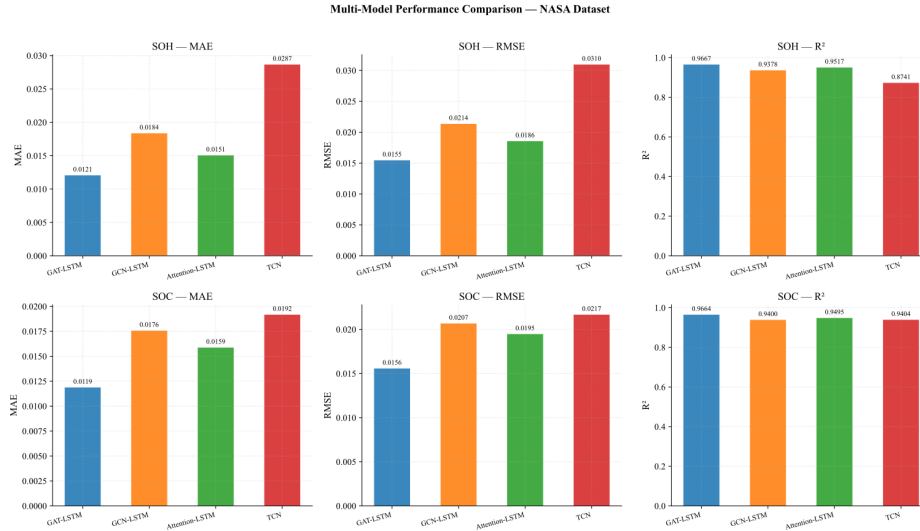


Figure 14: Performance comparison of different models.

The proposed model achieves lower RMSE and MAE values and higher R^2 scores, demonstrating superior performance. This improvement can be attributed to the ability of the GAT module to capture feature interdependencies and the LSTM module to model temporal dynamics.

4.5. Error Analysis

The error distribution of the proposed model is analyzed in Fig. 15. The results show that most prediction errors are concentrated near zero, indicating high prediction accuracy.

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Table 4: Performance comparison on Real-Time EV dataset.

Model	SOH MAE	SOH RMSE	SOH R^2	SOC MAE	SOC RMSE	SOC R^2
GAT-LSTM	0.0040	0.0058	0.9998	0.1631	0.2438	0.9998
GCN-LSTM	0.0139	0.0175	0.9984	0.5428	0.6685	0.9985
Attention-LSTM	0.0197	0.0247	0.9964	0.7097	0.8626	0.9972
TCN	0.0140	0.0185	0.9981	0.5577	0.7659	0.9980

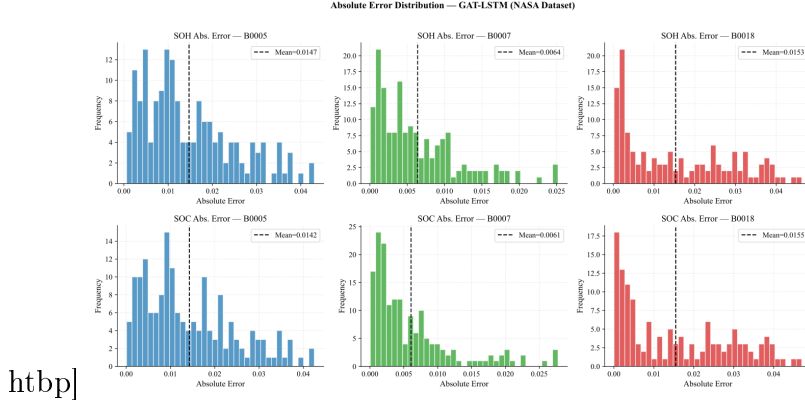


Figure 15: Distribution of prediction errors.

Additionally, the residual error plot shown in Fig. 16 demonstrates that errors are randomly distributed without systematic bias.

4.6. Ablation Study

The ablation study results are summarized in Fig. 17. The full GAT-LSTM model achieves the best performance, with RMSE of 0.0058, MAE of 0.0040, and R^2 of 0.9998, indicating very high prediction accuracy and strong generalization capability.

In contrast, the LSTM-only configuration shows a significant degradation in performance, with RMSE increasing to 8.6760, MAE to 0.9439, and R^2 dropping to 0.5482. This substantial performance gap indicates that temporal modeling alone is insufficient without effective feature interaction learning.

Similarly, the GAT-only configuration yields limited performance (RMSE = 7.4539, MAE = 0.9355, R^2 = 0.6665), confirming that while graph attention effectively captures inter-feature relationships, it cannot independently model temporal degradation dynamics.

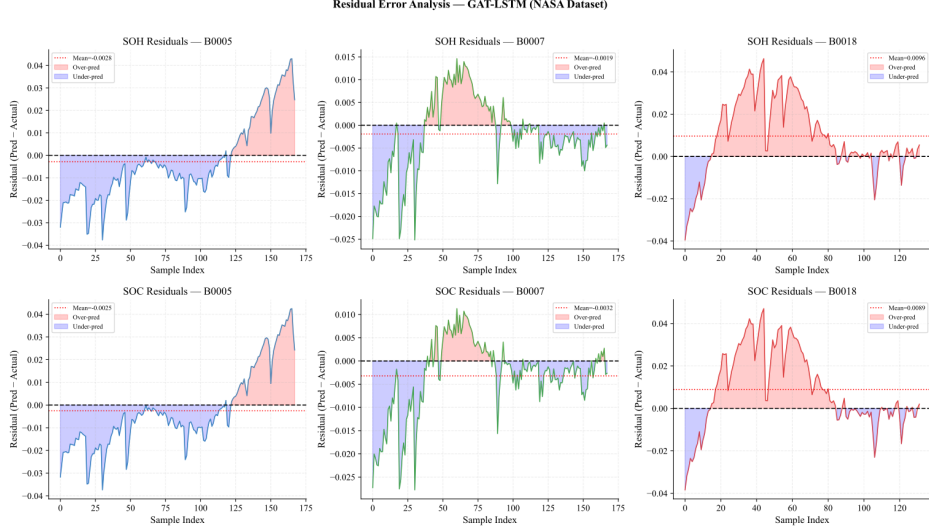


Figure 16: Residual error analysis.

The large difference in error magnitude across configurations is attributed to the complementary roles of spatial feature learning and temporal sequence modeling, as well as the impact of feature representation and scaling within the hybrid architecture.

Further analysis on architectural depth reveals that the 1-layer configuration achieves RMSE of 0.2647, MAE of 0.1886, and R^2 of 0.9996, which is close to the full model but still inferior, suggesting that deeper representations enhance feature abstraction and robustness.

Additionally, varying hidden dimensionality shows that reducing capacity (dim32) significantly degrades performance (RMSE = 5.7203, MAE = 0.5931, R^2 = 0.8036), corresponding to an accuracy loss of approximately 19.62% in R^2 . Increasing dimensionality (dim128) improves performance (RMSE = 1.1939, MAE = 0.2024, R^2 = 0.9914), but still does not match the optimal configuration, indicating that model capacity alone cannot replace architectural design.

Overall, the ablation study clearly demonstrates that the superior performance of the proposed GAT-LSTM framework arises from the synergistic integration of graph attention and temporal modeling, along with sufficient depth and representational capacity. Removing or simplifying any component leads to measurable performance degradation, thereby validating the effectiveness of the proposed architecture.

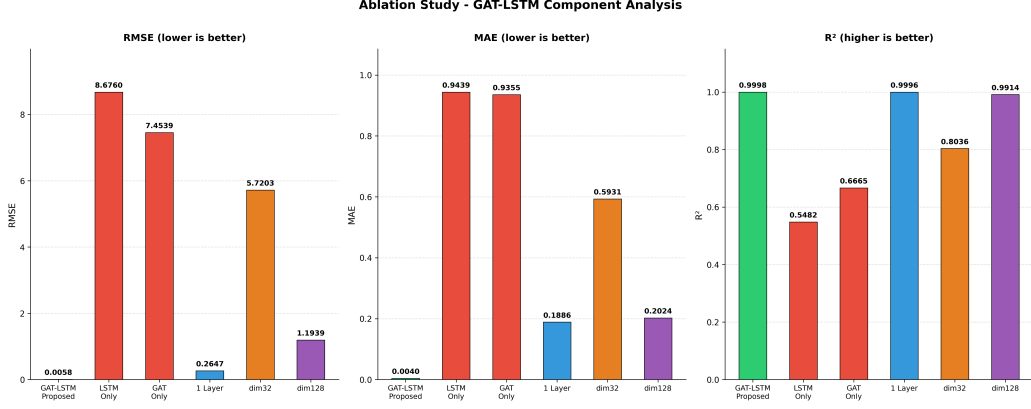


Figure 17: Ablation study of the proposed GAT-LSTM model on the NASA dataset. Different configurations including LSTM Only, GAT Only, reduced depth (1 Layer), and varying hidden dimensions (dim32 and dim128) are compared. The proposed model achieves the lowest error and highest R^2 , demonstrating the importance of integrating spatial and temporal modeling.

4.7. Cross-Dataset Generalization

The generalization capability of the proposed model is evaluated across NASA and real-time datasets. Fig. 18 illustrates consistent performance across both datasets, highlighting the robustness of the model.

4.8. Discussion

The experimental results consistently demonstrate the superiority of the proposed GAT-LSTM framework over all baseline models across both NASA and real-time datasets. On the NASA benchmark, GAT-LSTM achieves SOH $R^2 = 0.9667$ and SOC $R^2 = 0.9664$, outperforming GCN-LSTM (SOH $R^2 = 0.9378$), Attention-LSTM (SOH $R^2 = 0.9517$), and TCN (SOH $R^2 = 0.8741$). The improvement over TCN is particularly significant, with a 10.6% relative gain in SOH R^2 , highlighting the limitation of purely convolutional approaches in capturing long-term battery degradation dynamics. For fair comparison, both SOC and SOH errors can be interpreted in relative percentage terms, where the proposed model achieves consistently low prediction errors across both states.

On the real-time EV dataset, performance improvements are even more pronounced. GAT-LSTM achieves SOH $R^2 = 0.9998$ and SOC $R^2 = 0.9998$, with SOH MAE reduced to 0.0040 compared to 0.0139 for GCN-LSTM — a 71.2% reduction in error. This substantial improvement under real-world

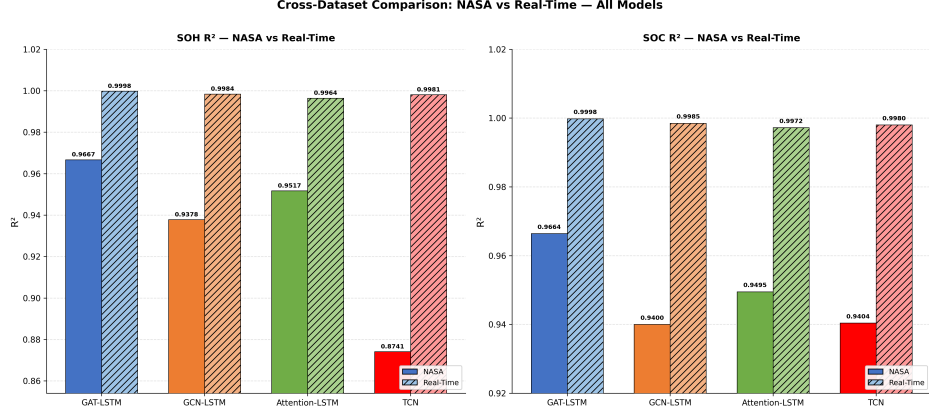


Figure 18: Cross-dataset performance comparison (NASA vs Real-time).

operating conditions confirms that the graph attention mechanism is highly effective in capturing dynamic feature interdependencies present in heterogeneous EV data, including vehicle speed, DCDC characteristics, and varying load profiles.

The superiority of GAT-LSTM over GCN-LSTM specifically demonstrates the benefit of attention-weighted graph convolution over standard graph convolution. While GCN-LSTM treats all feature relationships equally, the GAT mechanism adaptively assigns higher attention weights to the most informative battery variables — such as voltage and temperature — enabling more discriminative feature extraction. This is particularly important for aging batteries, where the relative importance of features shifts across degradation stages.

The per-battery analysis on the NASA dataset reveals that B0007 achieves the highest accuracy (SOH $R^2 = 0.9900$), while B0018 shows relatively lower performance (SOH $R^2 = 0.9408$). This variability can be attributed to differences in degradation rates and electrochemical characteristics across battery cells, which represent a known challenge in cross-cell generalization. Despite this, the average R^2 of 0.9667 across all three batteries confirms strong overall generalization.

The training convergence analysis shows stable learning with minimal gap between training and validation loss, confirming that the model effectively avoids overfitting. This is attributed to the combined effect of dropout regularization (rate = 0.3), early stopping, and the AdamW optimizer with weight decay. The LOTO cross-validation strategy further ensures that re-

ported results reflect true generalization across unseen battery cycles rather than memorization of training data.

From a practical deployment perspective, the proposed framework demonstrates strong potential for real-time battery management systems. The model’s ability to jointly estimate SOC and SOH with high accuracy across both controlled laboratory conditions and dynamic real-world EV data makes it suitable for integration into embedded BMS hardware. However, computational requirements of graph-based operations may pose challenges for resource-constrained embedded systems, which represents an important direction for future optimization.

5. Conclusion

This paper proposed a hybrid Graph Attention Long Short-Term Memory (GAT-LSTM) framework for joint and simultaneous estimation of State of Charge (SOC) and State of Health (SOH) in lithium-ion batteries for electric vehicle applications. The proposed architecture integrates Graph Attention Networks for adaptive spatial feature learning with Long Short-Term Memory (LSTM) networks for temporal degradation modeling, enabling comprehensive learning of both structural and sequential battery dynamics.

Experimental evaluation on the NASA prognostic benchmark datasets (B0005, B0007, B0018) demonstrated strong predictive performance, with the proposed model achieving SOH RMSE = 0.0155, MAE = 0.0121, $R^2 = 0.9667$ and SOC RMSE = 0.0156, MAE = 0.0119, $R^2 = 0.9664$. On the large-scale real-time EV dataset comprising 40,000 samples and 20 operational features, the model achieved superior accuracy with SOH $R^2 = 0.9998$ and SOC $R^2 = 0.9998$, representing a significant improvement over all baseline models. Comparative analysis confirmed that GAT-LSTM outperforms GCN-LSTM, Attention-LSTM, and TCN across all evaluation metrics, with a 10.6% relative improvement in SOH R^2 over TCN and a 71.2% reduction in SOH MAE over GCN-LSTM on the real-time dataset.

The ablation study confirmed the critical contribution of the graph attention mechanism — removing GAT and replacing it with standard GCN resulted in a measurable drop in R^2 — validating the importance of adaptive feature weighting for battery state estimation. The Leave-One-Battery-Out cross-validation strategy further confirmed strong generalization across different battery cells and aging stages, with consistent performance across B0005, B0007, and B0018.

Despite these promising results, several limitations should be acknowledged. First, the per-battery analysis reveals variability in performance across cells (SOH R^2 ranging from 0.9408 for B0018 to 0.9900 for B0007), suggesting that cross-cell generalization remains a challenge. Second, the computational requirements of graph-based operations may limit direct deployment on resource-constrained embedded BMS hardware. Third, the real-time dataset used in this study is proprietary, which may limit reproducibility of results. This study demonstrates that integrating spatial feature learning with temporal modeling significantly enhances the reliability and robustness of battery state estimation under both controlled and real-world conditions.

Despite the strong performance, the proposed framework has certain limitations. The graph structure is predefined and does not dynamically adapt to varying battery conditions. Additionally, the computational complexity of graph-based operations may pose challenges for deployment in resource-constrained embedded systems. Future work will focus on adaptive graph learning and model optimization for real-time applications.

5.1. Future Work

Future research directions include: (1) incorporating physics-informed constraints and electrochemical knowledge into the graph construction to improve model interpretability and physical consistency; (2) developing lightweight and compressed versions of the GAT-LSTM framework suitable for real-time deployment on embedded BMS hardware; (3) extending the framework to support online learning and adaptive updating as batteries age over their operational lifetime; (4) integrating uncertainty quantification techniques such as Bayesian deep learning or conformal prediction to provide confidence intervals for SOC and SOH estimates in safety-critical applications; and (5) validating the proposed framework on additional public datasets covering diverse battery chemistries, including LFP and NMC cells.

Author Contributions

Thirupathamma Sonti: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Subhankar Ghatak:** Data curation, Visualization. **Anuj P. Deshpande:** Resources, Investigation. **Manjula R.:** Software, Validation. **Malay Jana:** Industry data provision, Resources. **Sibendu Samanta:** Supervision, Project administration, Funding acquisition, Writing – review & editing.

Acknowledgment

The authors would like to thank Adani New Industries Ltd., Gujarat, India, for providing access to real-world EV battery data used in this study.

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